



Monster Elements headphone

Monster's new headphone Elements comes with a new music-centric voice assistant named Melody. <http://dgit.in/MonsterAssist>



A drone with claws

Researchers at the University of Sherbrooke have designed a new spying drone that can land on walls using claws. <http://dgit.in/DroneClaw>



Galileo with his telescope in the Piazza San Marco, Venice. Wood engraving.

Sir Isaac Newton, with his observations based on his self-designed reflecting telescope, published his *Philosophiæ Naturalis Principia Mathematica* in 1687 establishing, among other things, the laws of gravitation. These very laws have helped astronomers greatly in understanding the forces between celestial bodies. The next century saw amateur astronomers take astronomy to new levels and even laid down the foundations of spectroscopy.

SPECTROSCOPY AND SPECTROMETER

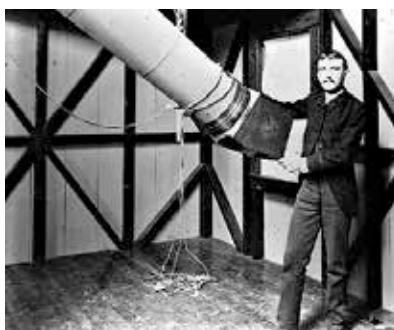
In an experiment to measure the energy given out by different colours that compose sunlight, William Herschel had split it through a prism. He observed a sudden increase in energy beyond the red end of the spectrum, discovering invisible infrared and laying the foundations of spectroscopy for years to come. The first accurate spectrometer was built by Joseph von Fraunhofer in 1814, which eventually led to the study of what stars are made of, based on the spectral radiation they emit into space.

Interesting fact: A new bright line in the sun's spectrum was observed in 1868 and was concluded to be an element unknown to Earth. British astronomer Norman Lockyer called it helium, which is Greek for the Sun. About 30 years later, Helium was discovered on Earth.

SCIENTIFIC ASTROPHOTOGRAPHY

While amateur astronomers and scientists realised the potential of photography as a tool for astronomical observation, there were technical challenges involved. These mostly included the lack of ways to keep a telescope accurately aimed at a fixed point over a long period of time. Additionally, early photographic processes like the daguerreotype had their own fair share of limitations.

The first real scientific astrophotograph was obtained by French physicists Jean Foucault and Armand Fizeau, as the picture of the Sun's surface through a telescope. Soon after that, better methods and technological advancements led to better and clearer pictures. A range of phenomena and celestial bodies like a solar eclipse, the moon and more, were captured, providing astronomers greater clarity in



Henry Draper with a refractor telescope set up for photography (photo probably taken in the 1860s or early 1870)

their study of the cosmos. For instance, post the 1872 photograph of Vega by Henry Draper that showed absorption lines, William Huggins used absorption lines to measure the redshift in stars, giving, for the first time, an indication to how fast stars are moving.

Today, amateur astrophotography is possible even on common film and digital cameras with basic equipment. There is a wide range of commercial equipment geared toward basic and advanced astrophotography, like single-lens reflex cameras, 35mm film, digital single-lens reflex cameras, and even off-the-shelf webcams adapted for long-exposure imaging. A deeper look into amateur astrophotography calls for a different article by itself.

MEGA TELESCOPES AND PROBES

Trying to list all astronomical innovations and their impact would be an 'astronomical' task. Rather, here are a couple of those that are, to this date, considered milestones in themselves:

- **Very Large Array:** An array of 27 radio antennas, each 25 metre across, in New Mexico that is used to study everything from black holes to planetary nebulae.
 - **Hubble Space Telescope:** Since its launch in April 1990, the Hubble telescope has been used to study everything from the expansion rate of the universe to dark energy.
- Another area where astronomy has been greatly helped by technology is with probes. And there are a few that are particularly significant.
- The Rosetta orbiter and the Philae lander that landed on comet 67P.
 - Voyager 1 and 2 have provided us with more detailed images of the gas giants than ever before.

INTERGALACTIC VISION

Astronomy has grown from being an observatory science to something that actually affects the future of humanity, as our quest to find an alternative planet to colonise draws nearer with each innovation in the space race. And if you want to be a part of this exploration, check out the DIY in this issue. 